3. If demand and the cost function of a firm are $p = 2 - x$ and $C = -2x^2 + 2x + 7$, then its profit function is
   (a) $x^2 + 7$ (b) $x^2 - 7$
   (c) $-x^2 + 7$ (d) $-x^2 - 7$

4. If the demand function is said to be elastic, then
   (a) $|\eta_d| > 1$ (b) $|\eta_d| = 1$
   (c) $|\eta_d| < 1$ (d) $|\eta_d| = 0$

5. The elasticity of demand for the demand function $x = \frac{1}{p}$ is
   (a) 0 (b) 1 (c) $-\frac{1}{p}$ (d) $\infty$

6. Relationship among $MR$, $AR$ and $\eta_d$ is
   (a) $\eta_d = \frac{AR}{AR - MR}$ (b) $\eta_d = AR - MR$
   (c) $MR = AR = \eta_d$ (d) $AR = \frac{MR}{\eta_d}$

7. For the cost function $C = \frac{1}{25}e^{5x}$, the marginal cost is
   (a) $\frac{1}{25}$
   (b) $\frac{1}{5}e^{5x}$
   (c) $\frac{1}{125}e^{5x}$
   (d) $25e^{5x}$

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8. Instantaneous rate of change of $y = 2x^2 + 5x$ with respect to $x$ at $x = 2$ is
   (a) 4 (b) 5 (c) 13 (d) 9

9. If the average revenue of a certain firm is ₹ 50 and its elasticity of demand is 2, then their marginal revenue is
   (a) ₹ 50 (b) ₹ 25 c) ₹ 100 (d) ₹ 75

10. Profit $P(x)$ is maximum when
    (a) $MR = MC$ (b) $MR = 0$
    (c) $MC = AC$ (d) $TR = AC$

11. The maximum value of $f(x) = \sin x$ is
    (a) 1 (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) $-\frac{1}{\sqrt{2}}$

12. If $f(x, y)$ is a homogeneous function of degree $n$, then $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y}$ is equal to
    (a) $(n-1)f$ (b) $n(n-1)f$
    (c) $nf$ (d) $f$

13. If $u = 4x^2 + 4xy + y^2 + 4x + 32y + 16$. then $\frac{\partial^2 u}{\partial y \partial x}$ is equal to
    (a) $8x + 4y + 4$ (b) 4
    (c) $2y + 32$ (d) 0

14. If $u = x^3 + 3xy^2 + y^3$, then $\frac{\partial^2 u}{\partial y \partial x}$ is
    (a) 3 (b) $6y$ (c) $6x$ (d) 2

15. If $u = e^{x^2}$, then $\frac{\partial u}{\partial x}$ is equal to
    (a) $2xe^{x^2}$ (b) $e^{x^2}$ (c) $2e^{x^2}$ (d) 0

16. Average cost is minimum when
    (a) Marginal cost = Marginal revenue
    (b) Average cost = Marginal cost
    (c) Average cost = Marginal revenue
    (d) Average Revenue = Marginal cost

17. A company begins to earn profit at
    (a) Maximum point
    (b) Breakeven point
    (c) Stationary point
    (d) Even point

18. The demand function is always
    (a) Increasing function
    (b) Decreasing function
    (c) Non-decreasing function
    (d) Undefined function

19. If $q = 1000 + 8p_1 - p_2$, then $\frac{\partial q}{\partial p_1}$ is
    (a) –1 (b) 8
    (c) 1000 (d) $1000 - p_2$



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